**RAFFLES INSTITUTION**
**H2 Mathematics (9758)**
**2026 Year 6**

# Additional Practice Questions for Chapter S6: Correlation and Regression

[**Note**: You only need to give the numerical value of the product moment correlation coefficient correct to 3 s.f. or the accuracy as required by the question. The answer in the solution listed the value to a higher degree of accuracy as a checkpoint for you to check that you have keyed in the data correctly.]

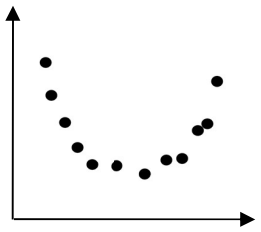
**1** Explain why it is advisable to draw a scatter diagram before interpreting a correlation coefficient calculated for a sample drawn from a bivariate distribution. [1]

Sketch a scatter diagram that shows an obvious relation between two variables but which yields a linear product-moment correlation coefficient that is close to zero. [1]

**Solution:**

A scatter diagram helps to assess if a linear relationship exists between the 2 variables, or a non-linear relationship fits better. It also helps to identify any outliers in the data that do not fit in the general pattern.

One possible scatter diagram is



**2** Seven cities in a certain country are linked by rail to the capital city. The table below shows the distance of each city from the capital and the rail fare from the city to the capital.

| City | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ | $G$ |
|---|---|---|---|---|---|---|---|
| Distance, $x$ km | 124 | 44 | 76 | 148 | 16 | 180 | 104 |
| Rail fare, \$ $y$ | 156 | 53 | 99 | 169 | 23 | 177 | 138 |

**(i)** Give a sketch of the scatter diagram for the data as shown on your calculator. [2]

**(ii)** Calculate the product moment correlation coefficient. [1]

You are given that the regression line of $y$ on $x$ has equation $y = 16.7 + 1.01x$, where the coefficients are given correct to 3 significant figures.

**(iii)** Calculate the equation of the regression line of $x$ on $y$, giving your answer in the form $x = a + by$. [1]

**(iv)** Use the appropriate regression line to estimate

**(a)** the rail fare from a city that is 28 km from the capital, [2]

**(b)** the distance of a city from the capital if the rail fare is \$198. [2]

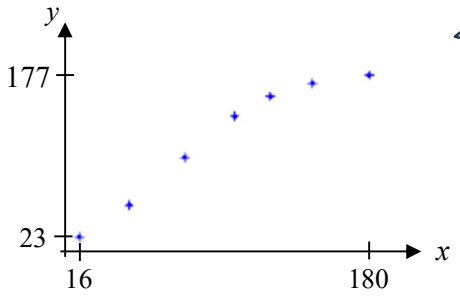
**(v)** Comment briefly on the reliability of the estimates in part **(iv)**. [2]

[**(ii)** $r = 0.973$ (3 s.f.) **(iii)** $x = -10.6 + 0.940y$ (3 s.f.) **(iv)(a)** \$45 **(b)** 176 km ]

**8863/2007/01/Q8**

**Solution:**

**(i)**




When drawing a scatter diagram:
(i) ensure that the relative positions of the points are correct
(ii) label the axes and range of the data

**(ii)** Using GC, $r \approx 0.97348 = 0.973$ (3 s.f.)

**(iii)** $x = -10.558 + 0.93976y$ (5 s.f.)
$x = -10.6 + 0.940y$ (3 s.f.)

| | |
|---|---|
| **(iv) (a)** | Given $x = 28$, use $y = 16.7 + 1.01x$ (regression line of $y$ on $x$)<br>$y = 44.98 = 45$ (nearest integer).<br>So the estimated rail fare is \$45. |
| **(iv) (b)** | Given $y = 198$, use $x = -10.558 + 0.93976y$ (regression line of $x$ on $y$)<br>$x = 176$ (3 s.f.)<br>So the estimated distance is 176 km. |
| **(v)** | The value $x = 28$ is **within** the given range of values of $x$ where $16 \le x \le 180$. Since **interpolation** was used, and the correlation coefficient 0.973 is close to 1, indicating a strong positive linear correlation between the variables $x$ and $y$, the estimate obtained in **(a)** is likely to be reliable.<br><br>The value $y = 198$ is **outside** the given range of values of $y$ where $23 \le y \le 177$. Since **extrapolation** was used, the estimate obtained in **(b)** is unlikely to be reliable as the linear model may not be applicable outside the given range of data. |

Take note of what is required when answering questions on reliability of estimates.

**3** An engineering company makes cranes. The numbers, $x$, sold in each three-month period for two years, together with the profits, $y$ thousand dollars, on the sale of these cranes are given in the following table.

| $x$ | 15 | 17 | 13 | 21 | 16 | 22 | 14 | 18 |
|---|---|---|---|---|---|---|---|---|
| $y$ | 290 | 350 | 270 | 430 | 340 | 410 | 300 | 360 |

**(i)** Give a sketch of the scatter diagram for the data as shown in your calculator. [2]

**(ii)** Find $\bar{x}$ and $\bar{y}$, and mark the point $(\bar{x}, \bar{y})$ on your scatter diagram. [2]

**(iii)** Calculate the equation of the regression line of $y$ on $x$, and draw this line on your scatter diagram. [2]

**(iv)** Calculate the product moment correlation, and comment on its value in relation to your scatter diagram. [2]

**(v)** For the next three-month period, the sales target is 20 cranes. Estimate the corresponding profit. [2]

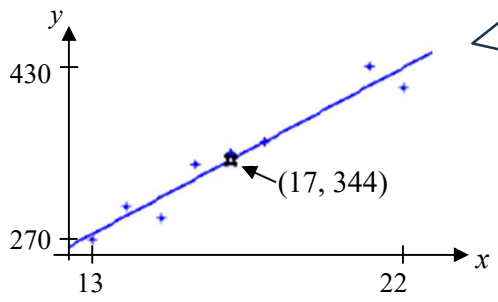
**(vi)** The company's sales director uses the regression line in **(iii)** to predict the profit if 40 cranes were to be sold in a three-month period. Comment on the validity of this prediction. [2]

[**(ii)** $\bar{x} = 17, \bar{y} = 344$ **(iii)** $y = 17.1x + 53.3$ **(iv)** 0.969 **(v)** \$395,000]

**8863/2008/01/Q11**

**Solution:**

**(i)**



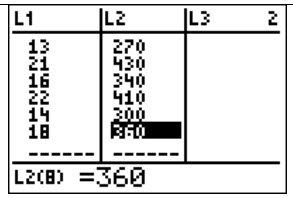

When drawing a scatter diagram:
(i) ensure that the relative positions of the points are correct
(ii) label the axes and range of the data

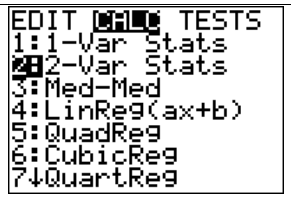
**(ii)**

$$\bar{x} = \frac{\sum x}{n} = 17$$

$$\bar{y} = \frac{\sum y}{n} = 343.75 = 344 \quad \text{(to nearest integer)}$$

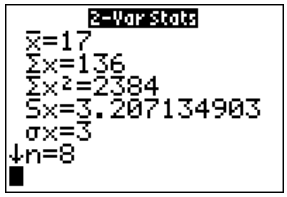
OR: use GC to find $\bar{x}$ and $\bar{y}$:

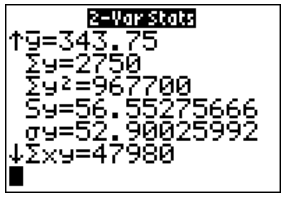






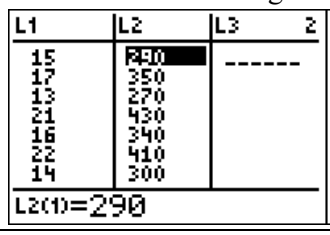



**(iii)** Linear regression of $y$ on $x$ is $y = 17.083x + 53.333$ (5 s.f.)

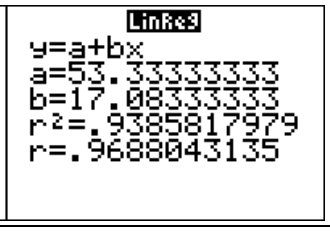
$$\Rightarrow y = 17.1x + 53.3 \text{ (3 s.f.)}$$

To use GC to draw regression line :










**(iv)** $r \approx 0.96880 = 0.969$ (3 s.f.)

The product moment correlation coefficient is close to 1, suggesting a strong positive linear correlation between $x$ and $y$. This is consistent with the scatter diagram which shows most of the points lying close to the regression line with a positive gradient.

**(v)** When $x = 20$, $y = 17.083(20) + 53.333 = 394.993$

i.e, the estimated corresponding profit is \$395 000.

**(vi)** Since $x = 40$ is **outside the given range of values of *x*** which is $[13, 22]$, **extrapolation** is used. The director's prediction would be unreliable as the linear model may not be applicable outside the given range of data.

**4** A medical officer wishes to investigate a patient's heart-beat rate $h$ beats per minute (bpm) when he walks at different speeds $s$ km/h. The data is shown below:

| $s$ (km/h) | 1 | 1.5 | 2 | 2.5 | 3 | 3.5 | 4 | 4.5 | 5 |
|---|---|---|---|---|---|---|---|---|---|
| $h$ (bpm) | 60 | 63 | 66 | 75 | 86 | 99 | 150 | 110 | 130 |

**(i)** Sketch a scatter plot of the above data. [1]

**(ii)** One of the values of $h$ appears to be incorrect. Indicate the corresponding point on your diagram by labelling it $P$. [1]

Omit $P$ for the remainder of this question.

**(iii)** Calculate the product moment correlation coefficient for this set of data. Use the equation of an appropriate regression line to predict the value of $s$ when $h = 100$, justifying your choice of regression line. [4]

It is suggested to use one of the following two models instead:

Model (I): $h = a + bs^2$,

Model (II): $h = a + b\mathrm{e}^s$,

where $a$ and $b$ are real constants.

**(iv)** Determine which of the two models is a better choice, giving a reason for your answer. [2]

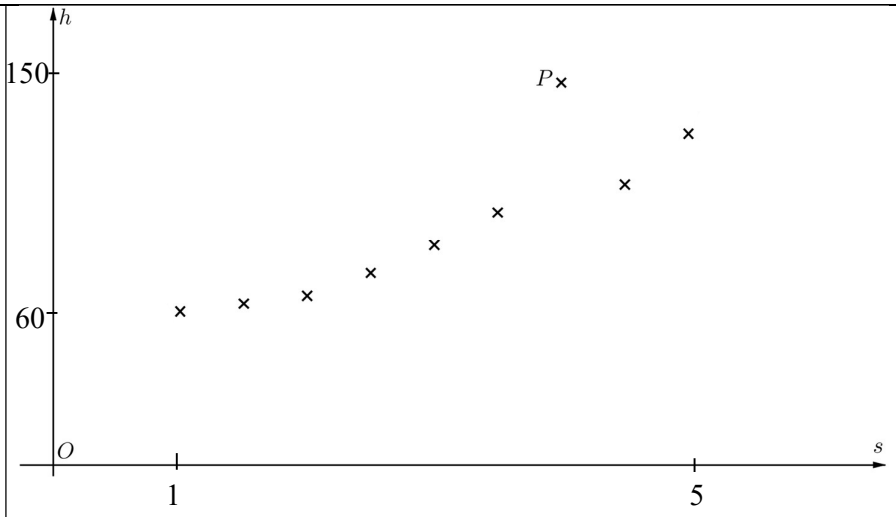
**(v)** Suppose a new data pair $(\bar{s}, \bar{h})$ is added to the table above, where $\bar{s}$ and $\bar{h}$ are the patient's sample mean walking speed (in km/h) and his sample mean heart-beat rate (in bpm) respectively, based on the data above. Without any calculations, explain whether the equation of the regression line you have obtained in part **(iii)** would change. [2]

[**(iii)** 0.981, $s = 3.67$ **(iv)** model (I)]

**9740/2015/NJC/02/Q12 (modified)**

**Solution:**

**(i)**
**(ii)**




**(iii)** $r$-value $\approx 0.98119 = 0.981$ (3 s.f.)

$h$ on $s$ line: $h = 35.8514 + 17.4865s$ (6 s.f.)
When $h = 100$, $s \approx 3.6685 = 3.67$ (3 s.f.)

**Note:** We use the $h$ on $s$ line since $s$ is the independent variable.

**(iv)** For model (I), $r$-value $\approx 0.98979 = 0.990$ (3 s.f.)
For model (II), $r$-value $\approx 0.93804 = 0.938$ (3 s.f.)

Hence model (I) is more suitable since the correlation coefficient is closer to 1, which suggests a stronger linear relationship between $h$ and $s^2$ as compared to that between $h$ and $e^s$.

**(v)** Since the data point $(\bar{s}, \bar{h})$ lies on the regression line, the answer to **(iii)** will not change.

**OR**

The addition of the data point $(\bar{s}, \bar{h})$ does not add further to the sum of squares of the vertical residuals between the current regression line and the data points, hence the sum of squares of vertical residuals remain minimised with the same line.

**5** **(a)** Sketch a scatter diagram that might be expected when $x$ and $y$ are related approximately by $y = px^2 + t$ in each of the cases **(i)** and **(ii)** below. In each case your diagram should include 6 points, approximately equally spaced with respect to $x$, and with all $x$-values positive.

**(i)** $p$ and $t$ are both positive.

**(ii)** $p$ is negative and $t$ is positive. [2]

**(b)** The ages in months ($m$) and prices in dollars ($P$) of a random sample of ten used cars of a certain model are given in the table.

| $m$ | 11 | 20 | 28 | 36 | 40 | 47 | 58 | 62 | 68 | 75 |
|---|---|---|---|---|---|---|---|---|---|---|
| $P$ | 112800 | 102600 | 76500 | 72000 | 72000 | 69000 | 65800 | 57000 | 50600 | 47600 |

It is thought that the price after $m$ months can be modelled by one of the formulae

$$P = am + b, \qquad P = c\ln m + d,$$

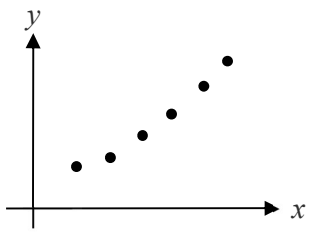
where $a$, $b$, $c$ and $d$ are constants.

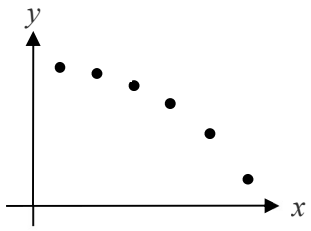
**(i)** Find, correct to 4 decimal places, the value of the product moment correlation coefficient between
**(A)** $m$ and $P$,
**(B)** $\ln m$ and $P$. [2]

**(ii)** Explain which of $P = am + b$ and $P = c\ln m + d$ is the better model and find the equation of a suitable regression line for this model. [3]

**(iii)** Use the equation of your regression line to estimate the price of a car that is 50 months old. [1]

[**(b)(i)** $-0.9470$, $-0.9749$ **(ii)** $P = 196000 - 33700\ln m$ **(iii)** \$64000]

| **9740/2014/02/Q8** | |
|---|---|
| **Solution:** | |
| **(a)** **(i)** |   |

| | |
|---|---|
| **(a)**<br>**(ii)** |  |
| **(b)**<br>**(i)** | Product moment correlation coefficient between $m$ and $P$ is –0.9470 (4 d.p.).<br><br>Product moment correlation coefficient between $\ln m$ and $P$ is –0.9749 (4 d.p.). |
| **(b)**<br>**(ii)** | –0.9749 is closer to –1 compared to –0.9470.<br>The scatter diagram of $P$ on $m$ also suggests a non-linear relationship between the 2 variables.<br>Thus $P = c\ln m + d$ is the better model.<br><br>$P = 195690 - 33660\ln m$ (6 s.f.)<br>$P = 196000 - 33700\ln m$ (3 s.f.) |
| **(b)**<br>**(iii)** | When $m = 50$, $P = 195690 - 33660\ln m$<br>$\approx 64011$<br>From GC, estimated price is \$64000 (3 s.f.) |

**6** Research is being carried out into how the concentration of a drug in the bloodstream varies with time, measured from when the drug is given. Observations of successive times give the data shown in the following table.

| Time ($t$ minutes) | 15 | 30 | 60 | 90 | 120 | 150 | 180 | 240 | 300 |
|---|---|---|---|---|---|---|---|---|---|
| Concentration ($x$ micrograms per litre) | 82 | 65 | 43 | 37 | 22 | 19 | 12 | 6 | 2 |

It is given that the value of the product moment correlation coefficient for this data is $-0.912$, correct to 3 decimal places. Draw a scatter diagram for the data. [1]

Calculate the equation of the regression line of $x$ on $t$. [2]

Calculate the corresponding estimated value of $x$ when $t = 300$, and comment on the suitability of the linear model. [2]

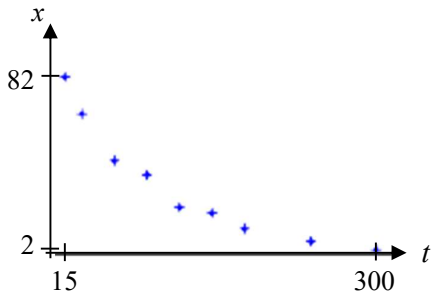
The variable $y$ is defined by $y = \ln x$. For the variables $y$ and $t$,

**(i)** calculate the product moment correlation coefficient and comment on its value, [2]

**(ii)** calculate the equation of the appropriate regression line. [3]

Use a regression line to give the best estimate that you can of the time when the drug concentration is 15 micrograms per litre. [2]

[$x = 66.2 - 0.260t$, $-11.7$ **(i)** $r = -0.994$ **(ii)** $y = 4.62 - 0.0123t$, 155]

| **9740/2007/02/Q11** | |
|---|---|
| **Solution:** | |
| |  |
| | Equation of regression line of $x$ on $t$ :<br>$x = 66.194 - 0.25970t$ (5 s.f.)<br>$x = 66.2 - 0.260t$ (3 s.f.) |

| | |
|---|---|
| | When $t = 300$, $x = 66.194 - 0.25970(300) = -11.7$ (3 s.f.)<br><br>Even though $r \approx -0.912$ indicates a strong negative linear correlation between the variables, $x$ and $t$, the linear model may not be suitable as the scatter diagram shows that the points follow a curved nature.<br>Also, it is not possible to have $x = -11.7$ as concentration of blood cannot be negative. |
| **(i)** | $r = -0.994$ (3 s.f.)<br>The $r$ obtained is closer to $-1$ as compared to the case when using variables $x$ and $t$. |
| **(ii)** | $y = 4.6206 - 0.012343t$ (5 s.f.)<br>$y = 4.62 - 0.0123t$ (3 s.f.) |
| | When $x = 15$, $y = \ln 15$,<br>$\therefore \ln 15 = 4.6206 - 0.012343t$<br>$\Rightarrow t = 155$ (3 s.f.) |

**7** Mr. Lee recorded the length of time, $y$ minutes, taken to travel to work when leaving home $x$ minutes after 6.00 am on seven selected mornings. The results are as follows:

| $x$ /min | 0 | 10 | 20 | 30 | 40 | 50 | 60 |
|---|---|---|---|---|---|---|---|
| $y$ /min | 16 | 27 | 28 | 39 | 43 | 48 | 51 |

**(i)** Calculate the equations of the least squares regression lines of $y$ on $x$ and $x$ on $y$. [2]

**(ii)** Mr. Lee forgot to add an observation for a particular morning but remembered that the two regression lines remain unchanged on its inclusion. Find this missing observation. [1]

Mr. Lee needs to report to work by 8.00 am. He leaves home $x$ minutes after 6.00 am and he arrives at his workplace $z$ minutes before 8.00 am.

**(iii)** Find $z$ in terms of $x$. [1]

**(iv)** Hence estimate, to the nearest minutes, the latest time Mr. Lee can leave home without arriving late at work. [2]

[**(i)** $y = 0.579x + 18.6,\ x = 1.67y - 30$ **(ii)** (30,36) **(iii)** $z = 101 - 1.58x$ **(iv)** 7.04 am]

| **AJC/2010/MidYr/02/Q5** | |
|---|---|
| **Solution:** | |
| **(i)** | The least squares regression line $y$ on $x$ is<br>$y = 0.57857x + 18.643$ (5 s.f.)<br>$y = 0.579x + 18.6$ (3 s.f.).<br><br>The least squares regression lines $x$ on $y$ is<br>$x = 1.6667y - 30$ (5 s.f.)<br>$x = 1.67y - 30$ (3 s.f.). (Note: 30 is an exact value) |
| **(ii)** | Since the two regression lines remained unchanged, the missing observation is $(\overline{x}, \overline{y})$ which lies on both regression lines.<br>,ie. $(\overline{x}, \overline{y}) = (30,36)$, the missing observation is (30,36). |
| **(iii)** | $x + y + z = 120 \Rightarrow z = 120 - x - y$<br><br>$\therefore\ z = (120 - x) - (0.57857x + 18.643)$<br>$= 101.357 - 1.57857x$<br>$= 101 - 1.58x$ ( 3 s.f.) |
| **(iv)** | When $z \geq 0, x \leq \dfrac{101.357}{1.57857} = 64.2$ (3 s.f.)<br>$\therefore$ Latest time is 7.04 am (nearest minute) |

**8** A website about electric motors gives information about the percentage efficiency of motors depending on their power, measured in horsepower. Xian has copied the following table for a particular type of electric motor, but he has copied one of the efficiency values wrongly.

| Power, $x$ | 1 | 1.5 | 2 | 3 | 5 | 7.5 | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Efficiency, $y$% | 72.5 | 82.5 | 84.0 | 87.4 | 87.5 | 88.5 | 89.5 | 90.2 | 91.0 | 91.7 | 92.4 |

**(i)** Plot a scatter diagram on graph paper for these values, labeling the axes, using a scale of 2 cm to represent 10% efficiency on the $y$-axis and an appropriate scale for the $x$-axis. On your diagram, circle the point that Xian has copied wrongly. [2]

For parts **(ii)**, **(iii)** and **(iv)** of this question you should **exclude** the point for which Xian has copied the efficiency value wrongly.

**(ii)** Explain from your scatter diagram why the relationship between $x$ and $y$ should not be modelled by an equation of the form $y = ax + b$. [1]

**(iii)** Suppose that the relationship between $x$ and $y$ are modelled by an equation of the form $y = \frac{c}{x} + d$, where $c$ and $d$ are constants. State with a reason whether each of $c$ and $d$ is positive or negative. [2]

**(iv)** Find the product moment correlation coefficient and the constants $c$ and $d$ for the model in part **(iii)**. [3]

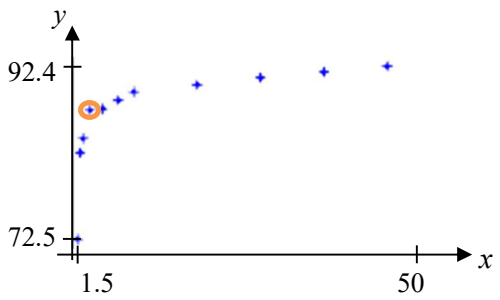
**(v)** Use the model $y = \frac{c}{x} + d$ with the values of $c$ and $d$ found in part **(iv)**, to estimate the efficiency value ($y$) that Xian has copied wrongly. Give two reasons why you would expect this estimate to be reliable. [3]

[**(iv)** $-0.980$, $c = -17.5$ and $d = 91.8$ **(v)** 85.9]

**9740/2016/02/Q8**

**Solution:**

**(i)**




**Note:**
Remember to use graph paper, and circle the point (3, 87.4) whose $y$-value is wrong.

**(ii)**

Scatter diagram indicates that **as $x$ increases**, $y$ **increases at a decreasing rate** and hence the relationship between $x$ and $y$ is non-linear. Thus it should not be modelled by an equation of the form $y = ax + b$.

Note how we describe a non-linear relationship between 2 variables.

**(iii)**

$$y = \frac{c}{x} + d$$

All $x$ and $y$ values are positive, thus $d > 0$ to account for large values of $x$ when $\frac{c}{x} \to 0$.

Values of $y$ increases as $x$ increase, so $c < 0$.

**(iv)**

From GC,
product moment correlation coefficient $\approx -0.97955 = -0.980$ (3 s.f.).

$$y = \frac{-17.4836}{x} + 91.7501 \quad (6 \text{ s.f.})$$

$c = -17.5$ (3sf) and $d = 91.8$ (3sf)

**(v)**

When $x = 3$, $y = \frac{-17.4836}{x} + 91.7501$

$\approx 85.922 = 85.9$ (3 s.f.)

We expect this estimate to be reliable as $r \approx -0.97955$ is close to $-1$ which suggests a strong negative linear correlation between the variables $\frac{1}{x}$ and $y$. In addition, the value $x = 3$ is within the given range of values of $x$ which is [1, 50], so we did not use extrapolation to obtain the estimate.

**9** For 3 years a factory records the number of units it produces each quarter and the total cost of producing the units. The following table summarises these results.

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Units of output ($x$) in 1000's | 15 | 30 | 57 | 75 | 10 | 25 | 49 | 70 | 20 | 34 | 60 | 76 |
| Total cost ($y$) in £1000's | 33 | 49 | 71 | 99 | 34 | 40 | 61 | 82 | 40 | 51 | 84 | 93 |

**(i)** Give a sketch of the scatter diagram for the data as displayed on your graphing calculator. [2]

**(ii)** Calculate the equation of the regression line of $y$ on $x$, draw this line on your diagram and calculate the product moment correlation coefficient. [3]

The selling price of each unit of output is £1.80.

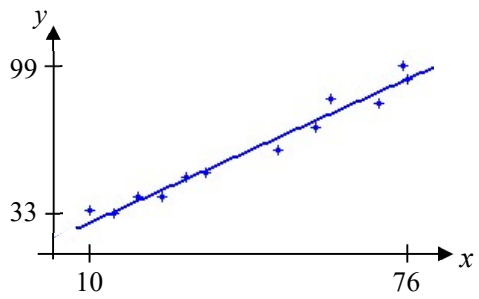
**(iii)** Use your graph to estimate the level of output (correct to 2 significant figures) at which the total cost and the total income balance. [2]

**(iv)** Give a brief interpretation of what this value means. [1]

[**(ii)** $y = 19.3 + 0.970x$, $r = 0.983$, **(iii)** 23000 units]

**Solution:**

**(i)**




**(ii)** From GC, $y = 19.281 + 0.97049x$ (5 s.f.)

$y = 19.3 + 0.970x$ (3 s.f.)

$r \approx 0.98314 = 0.983$ (3 s.f.)

**(iii)** Consider the intersection of the regression line and the line $y = 1.8x$ which gives $x \approx 23.244$

The level of output required is 23000 units (2 s.f.)

**(iv)** The factory makes a profit if the output is above this level. If the output is below this level, it suffers a loss.

**10** In an experiment the following information was gathered about air pressure $P$, measured in inches of mercury, at different heights above sea-level $h$, measured in feet.

| $h$ | 2000 | 5000 | 10000 | 15000 | 20000 | 25000 | 30000 | 35000 | 40000 | 45000 |
|---|---|---|---|---|---|---|---|---|---|---|
| $P$ | 27.8 | 24.9 | 20.6 | 16.9 | 13.8 | 11.1 | 8.89 | 7.04 | 5.52 | 4.28 |

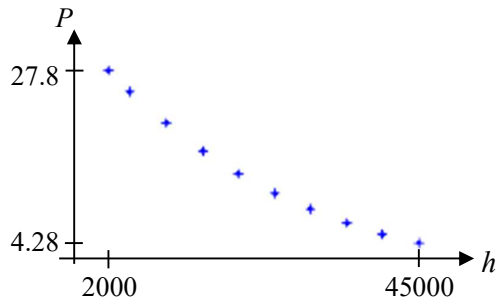
**(i)** Draw a scatter diagram for these values, labelling the axes. [1]

**(ii)** Find, correct to 4 decimal places, the product moment correlation coefficient between
**(a)** $h$ and $P$,
**(b)** $\ln h$ and $P$,
**(c)** $\sqrt{h}$ and $P$. [3]

**(iii)** Using the most appropriate case from part **(ii)**, find the equation which best models air pressure at different heights. [3]

**(iv)** Given that 1 metre = 3.28 feet, re-write your equation from part **(iii)** so that it can be used to estimate the air pressure when the height is given in metres. [2]

[**(ii)** $-0.9807,\ -0.9748,\ -0.9986$; **(iii)** $P = 34.8 - 0.147\sqrt{h}$ ; **(iv)** $P = 34.8 - 0.266\sqrt{h}$ ]

| **9740/2015/02/Q10** | |
|---|---|
| **Solution:** | |
| **(i)** |  |
| **(ii)** | From GC,<br>product moment correlation coefficient between $h$ and $P \approx -0.980731 = -0.9807$ ( 4 d.p.)<br>product moment correlation coefficient between $\ln h$ and $P \approx -0.9747996 = -0.9748$ ( 4 d.p.)<br>product moment correlation coefficient between $\sqrt{h}$ and $P \approx -0.998638 = -0.9986$ ( 4 d.p.) |
| **(iii)** | **(c)** has product moment correlation coefficient closest to –1. Equation of best model is<br>$P = 34.789 - 0.14687\sqrt{h}$ (5 s.f.)<br>$P = 34.8 - 0.147\sqrt{h}$ (3 s.f.) |
| **(iv)** | If height in metres is $h$, then height in feet is $3.28h$.<br>Then $P = 34.789 - 0.14687\sqrt{3.28h} \Rightarrow P = 34.8 - 0.266\sqrt{h}$ (3sf) |

**11 (i)** Sketch a scatter diagram that might be expected when $x$ and $y$ are related approximately as given in each of the cases (A), (B) and (C) below. In each case your diagram should include 6 points approximately equally spaced with respect to $x$, and with all $x$- and $y$-values positive. The letters $a$, $b$, $c$, $d$, $e$ and $f$ represent constants.

(A) $y = a + bx^2$, where $a$ is positive and $b$ is negative,

(B) $y = c + d\ln x$, where $c$ is positive and $d$ is negative,

(C) $y = e + \dfrac{f}{x}$, where $e$ is positive and $f$ is negative. [3]

A motoring website gives the following about the distance travelled, $y$ km, by a certain type of car at different speeds, $x$ kmh$^{-1}$, on a fixed amount of fuel.

| Speed, $x$ | 88 | 96 | 104 | 112 | 120 | 128 |
|---|---|---|---|---|---|---|
| Distance, $y$ | 148 | 147 | 144 | 138 | 126 | 107 |

**(ii)** Draw the scatter diagram for these values, labelling the axes. [1]

**(iii)** Explain which of the three cases in part **(i)** is the most appropriate for modelling these values, and calculate the product moment correlation coefficient for this case. [2]

**(iv)** It is required to estimate the distance travelled at a speed of 110 kmh$^{-1}$. Use the case that you identified in part **(iii)** to find the equation of a suitable regression line, and use your equation to find the required estimate. [3]

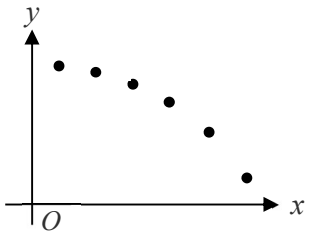
[**(iii)** model (A), $-0.939$]

**9740/2013/02/Q10**

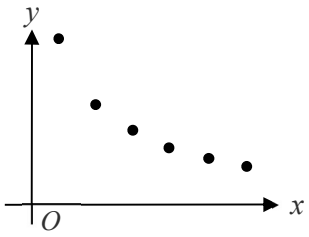
**Solution:**

**(i)**

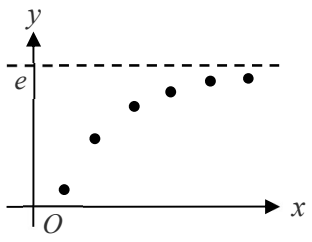
For (A): $y = a + bx^2$, with $a > 0$ and $b < 0$.



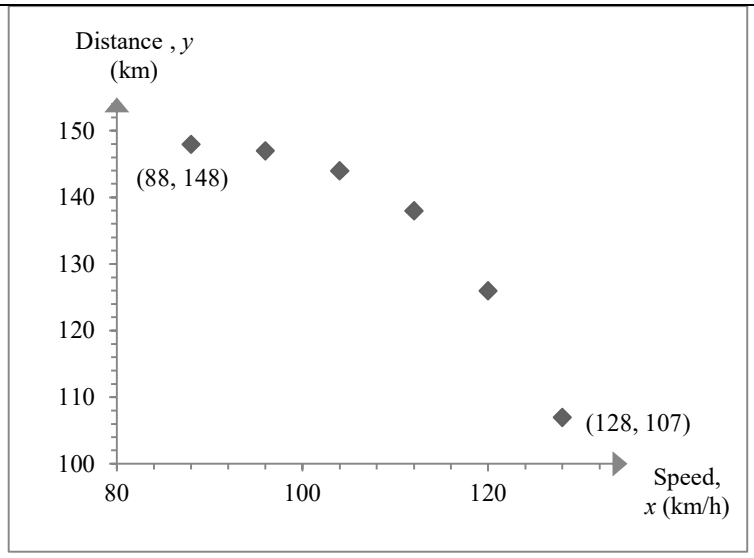

For (B): $y = c + d\ln x$, with $c > 0$ and $d < 0$.



For (C): $y = e + \dfrac{f}{x}$, with $e > 0$ and $f < 0$.




**(ii)**




**(iii)** From the scatter diagram, we see that **as $x$ increases**, **$y$ decreases at an increasing rate** which is consistent with the scatter plot for model (A). Thus (A) is the most appropriate model.
Using model (A), product moment correlation coefficient $\approx -0.93920 = -0.939$ (3 s.f.).

**(iv)** The required equation is $y = 189.7475 - 0.004619785x^2$ (7 s.f.)

$$\Rightarrow y = 190 - 0.00462x^2 \text{ (3 s.f.)}$$

At a speed of 110 km/h, the distance travelled is estimated to be
$y = 189.7475 - 0.004619785(110)^2 = 134$ km (3 s.f.).

**12** A car is placed in a wind tunnel and the drag force $F$ for different wind speeds $v$, in appropriate units is recorded. The results are shown in the table.

| $v$ | 0 | 4 | 8 | 12 | 16 | 20 | 24 | 28 | 32 | 36 |
|---|---|---|---|---|---|---|---|---|---|---|
| $F$ | 0 | 2.5 | 5.1 | 8.8 | 11.2 | 13.6 | 17.6 | 22.0 | 27.8 | 33.9 |

**(i)** Draw the scatter diagram for these values, labelling the axes clearly. [2]

It is thought that the drag force $F$ can be modelled by one of the formulae

$$F = a + bv \quad \text{or} \quad F = c + dv^2$$

where $a$, $b$, $c$ and $d$ are constants.

**(ii)** Find, correct to 4 decimal places, the value of the product moment correlation coefficient between

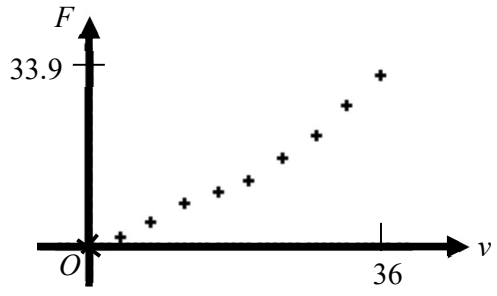
**(a)** $v$ and $F$,

**(b)** $v^2$ and $F$. [2]

**(iii)** Use your answers to parts **(i)** and **(ii)** to explain which of $F = a + bv$ or $F = c + dv^2$ is the better model. [1]

**(iv)** It is required to estimate the value of $v$ for which $F = 26.0$. Find the equation of a suitable regression line, and use it to find the required estimate. Explain why neither the regression line of $v$ on $F$ nor the regression line of $v^2$ on $F$ should be used. [4]

[**(ii)** 0.9860, 0.9907]

| **9740/2010/02/Q10** | |
|---|---|
| **Solution:** | |
| **(i)** |  Note that since $v$ is the independent variable, it must be on the horizontal axis! |
| **(ii)** | product moment correlation coefficient between $v$ and $F \approx 0.986024 = 0.9860$ (4 d.p.)<br>product moment correlation coefficient between $v^2$ and $F \approx 0.990681 = 0.9907$ (4 d.p.) |
| **(iii)** | Since the product moment correlation coefficient between $v^2$ and $F$ is closer to 1 as compared to between $v$ and $F$, $F = c + dv^2$ is the better model. The scatter diagram also |

| | |
|---|---|
| | shows a curvilinear trend (**as $v$ increases**, $F$ **increases at an increasing rate**), indicating that a linear model, $F = a + bv$ might not be appropriate. |
| **(iv)** | To estimate the value of $v$ for which $F = 26.0$, a suitable regression line is $F$ on $v^2$.<br><br>Using GC, $F = 3.19565 + 0.0242420v^2$ (6 s.f.)<br>When $F = 26.0$, $v \approx 30.671 = 30.7$ units (3 s.f.)<br><br>As $v$ is the independent variable, neither the regression line of $v$ on $F$ nor the regression line of $v^2$ on $F$ should be used. |

**13** In a study to determine the relationship between the temperature $T$ (measured in degrees Celsius) and the altitude $H$ (height above sea level, measured in metres) for a certain mountain, a team of meteorologists ascended and descended the mountain on a day with clear weather and obtained readings of $T$ and $H$.

The table below shows the 10 readings obtained:

| $H$ | 2300 | 2390 | 2626 | 2631 | 2700 | 3126 | 3131 | 3400 | 3450 | 3580 |
|---|---|---|---|---|---|---|---|---|---|---|
| $T$ | 11.80 | 9.26 | 10.29 | 7.99 | 9.52 | 5.20 | 7.26 | 3.72 | 4.43 | 2.31 |

**(a)** On the same axes, sketch a scatter diagram of the data and draw the regression line of $T$ against $H$. [2]

**(b)** State the product moment correlation coefficient between $T$ and $H$. Comment, in context, on the value obtained. [2]

It was found that 5 of these 10 readings were recorded between 2 am and 5.30 am when the team of meteorologists were ascending the mountain, while the other 5 were recorded between 8 am and 11.30 am when they were descending. The routes used for ascending and descending were the same. Between 5.30 am and 8 am, the meteorologists remained on the summit.

For these readings, the regression equation for ascending, corrected to 3 significant figures, is $T = 23.1 - 0.00574H$. The regression equation for descending is $T = a + bH$.

**(c)** Find, correct to 3 significant figures, the values of $a$ and $b$. [2]

One of the meteorologists recorded the temperature near the foot of the mountain as 26 degrees Celsius.

**(d)** Determine whether it is suitable to use any of the regression equations above to estimate the altitude of the foot of the mountain. [1]

A temperature of $F$ degrees Fahrenheit is equivalent to a temperature of $T$ degrees Celsius, where $F = \frac{9}{5}T + 32$.

**(e)** The regression equation for ascending, $T = 23.1 - 0.00574H$ can be written in the form $F = c + dH$, where $c$ and $d$ are constants. Find the values of $c$ and $d$. [2]

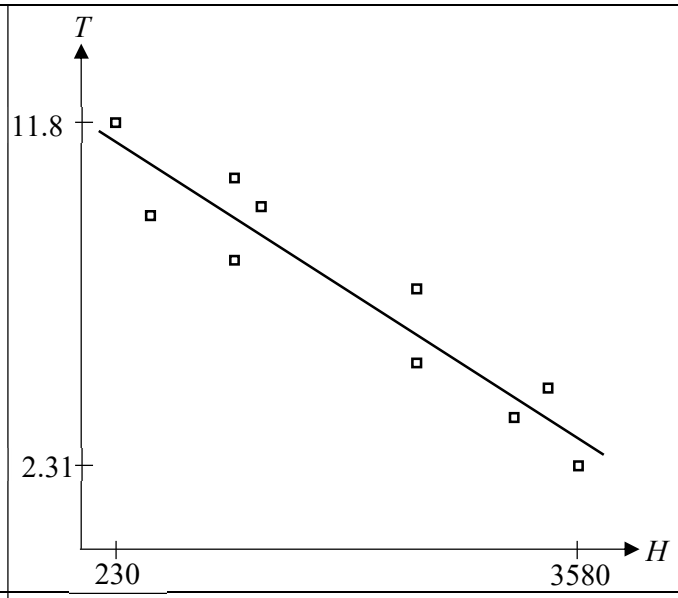
**(f)** Use the equation found in part **(e)** to estimate the temperature in degrees Fahrenheit when the meteorologists were at an altitude of 2800 m while ascending. Give two reasons why you would expect this estimate to be reliable. [3]

[**(b)** $r = -0.948$ **(c)** $a = 26.7$, $b = -0.00633$ **(e)** $c = 73.58$, $d = -0.010332$ **(f)** $F = 44.6504$]

**RI Prelim 9758/2023/02/Q9**

**Solution:**

**(a)**




**(b)** The product moment correlation coefficient between $T$ and $H$ is
$r \approx 0.94792 = -0.948$ (3 s.f.).

Since $r$ is close to $-1$, there is a strong negative linear correlation between $T$ and $H$. The temperature decreases as the altitude increases.

**(c)** The readings taken during descending are

| | | | | | |
|---|---|---|---|---|---|
| $H$ | 2300 | 2626 | 2700 | 3131 | 3450 |
| $T$ | 11.80 | 10.29 | 9.52 | 7.26 | 4.43 |

From GC, $a \approx 26.651 = 26.7$ (3 s.f.), $b \approx -0.0063317 = -0.00633$ (3 s.f.).

**(d)** None of the regression lines is suitable as 26 degree Celsius is outside the data range for $T$ where $2.31 \le T \le 11.80$. Extrapolation will be needed to obtain the estimate but the model may not be applicable outside the given range of data.

**(e)**
$$F = \frac{9}{5}(23.1 - 0.00574H) + 32$$
$$= 73.58 - 0.010332H$$

$\therefore c = 73.58$ and $d = -0.010332$.

| | |
|---|---|
| **(f)** | When $H = 2800$, $F = 73.58 - 0.010332(2800) = 44.6504$<br>The estimate is reliable because the value $H = 2800$ is within the data range of $2390 \le H \le 3580$ and the product moment correlation coefficient $r = -0.999$ is close to $-1$ indicating a strong negative linear correlation between $F$ and $H$ which suggests that the linear model is a suitable model. |

***THE END***